Power over Ethernet (PoE)

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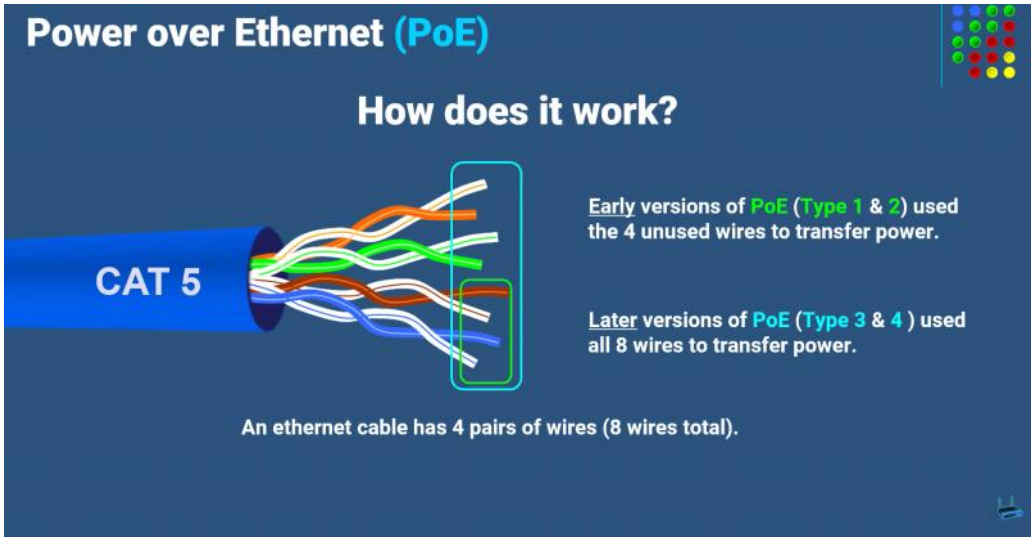
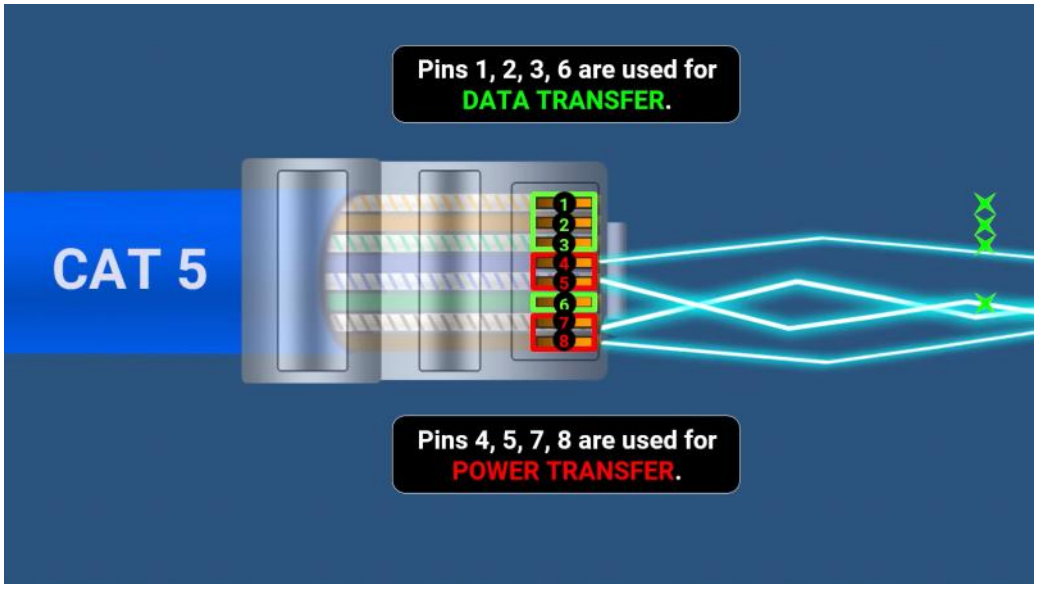
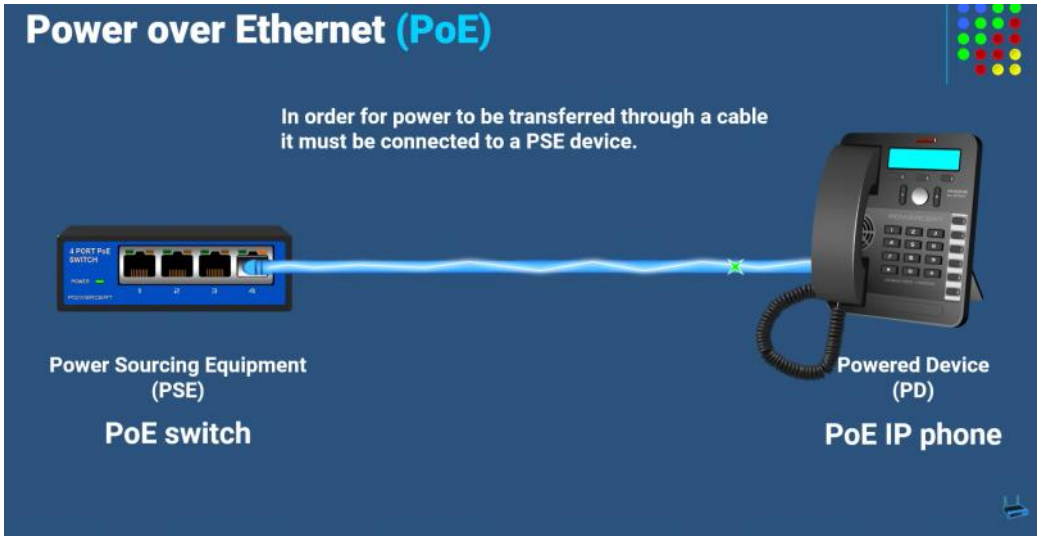
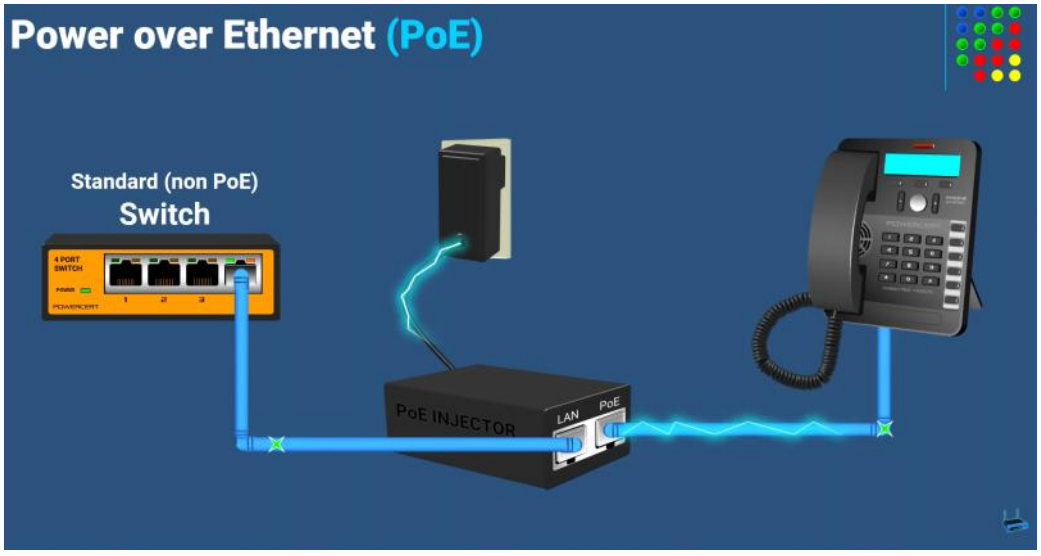
Power over Ethernet (PoE) is a technology that allows networking devices, such as IP phones, security cameras, and wireless access points, to receive both data and electrical power through a single Ethernet cable. This eliminates the need for separate power cables for each device.

How it Works:
An Ethernet cable contains four pairs of twisted wires. While some wires handle data transmission, others are used to deliver DC power. Systems utilizing this technology consist of a Power Sourcing Equipment (PSE), like a PoE switch, and a Powered Device (PD). If a standard non-PoE switch is in use, a PoE injector can be added to combine data from the switch and power from an outlet into a single cable.



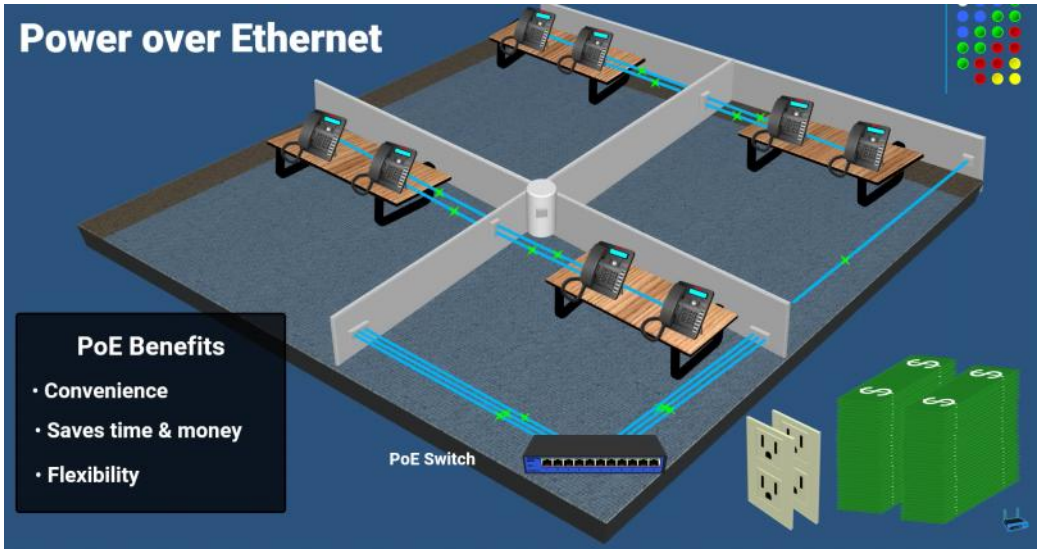






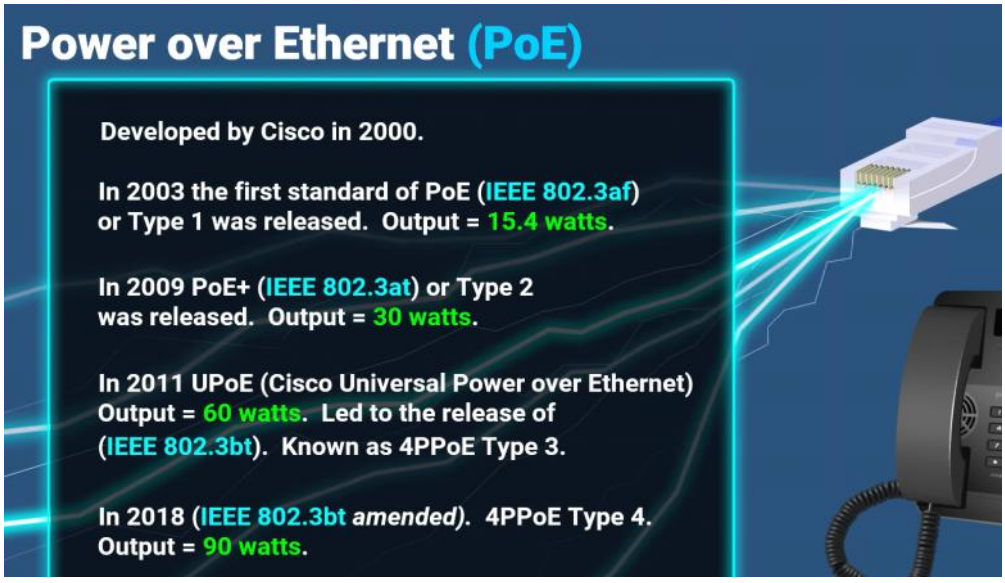
Key Benefits:

- Convenience: Reduces cable clutter.
- Cost-Efficiency: Eliminates the need to hire electricians to install additional power outlets.
- Flexibility: Allows devices to be placed in optimal locations regardless of proximity to power sources.



Types of PoE:

- Active PoE: The switch communicates with the device to verify compatibility and determine the correct power requirements before sending electricity, preventing damage.
- Passive PoE: Constantly sends power without communicating with the device. Caution is required to match voltage requirements, as incorrect connections can permanently damage hardware.

Standardization:

- PoE was developed by Cisco in 2000.
- PoE standards have evolved over time to support increasingly power-hungry devices, growing from the initial 15.4 Watts standard in 2003 to the modern 90 Watts standard, which is powerful enough to support devices like laptop computers.